

Clustering

Discovering Athlete Performance Profiles

Dataset: Athlete Performance Dataset | 1,000 rows | 20 Variables

Dataset Overview & Preprocessing

1,000

Observations

17

Features Used

3

Columns Dropped

3

Algorithms Tested

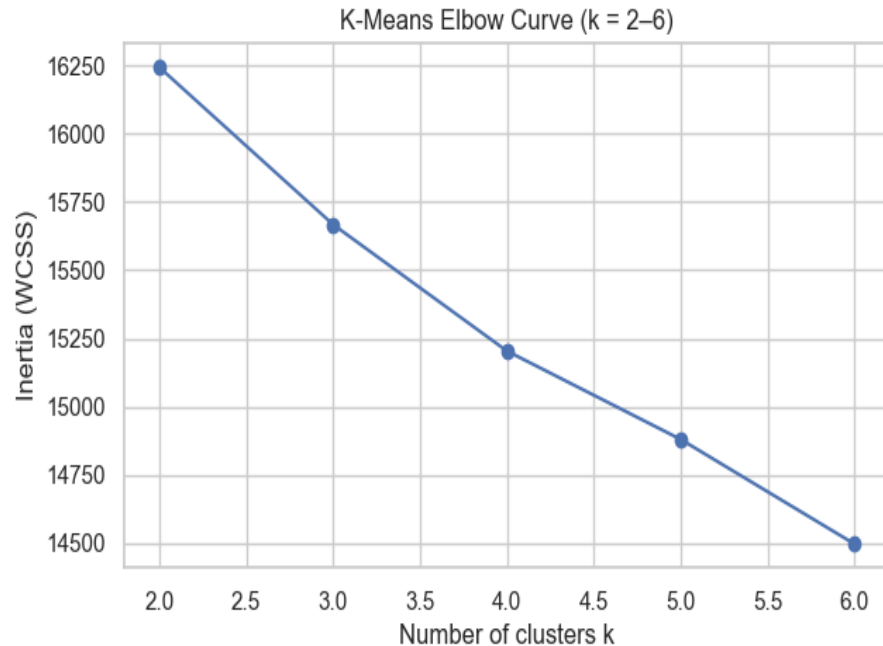
Preprocessing Steps

1. Dropped timestamp, performance_score, performance_category
2. Applied StandardScaler to all 17 numeric features
3. PCA to 2 components (visualisation only)
4. Clustering performed on full scaled feature set

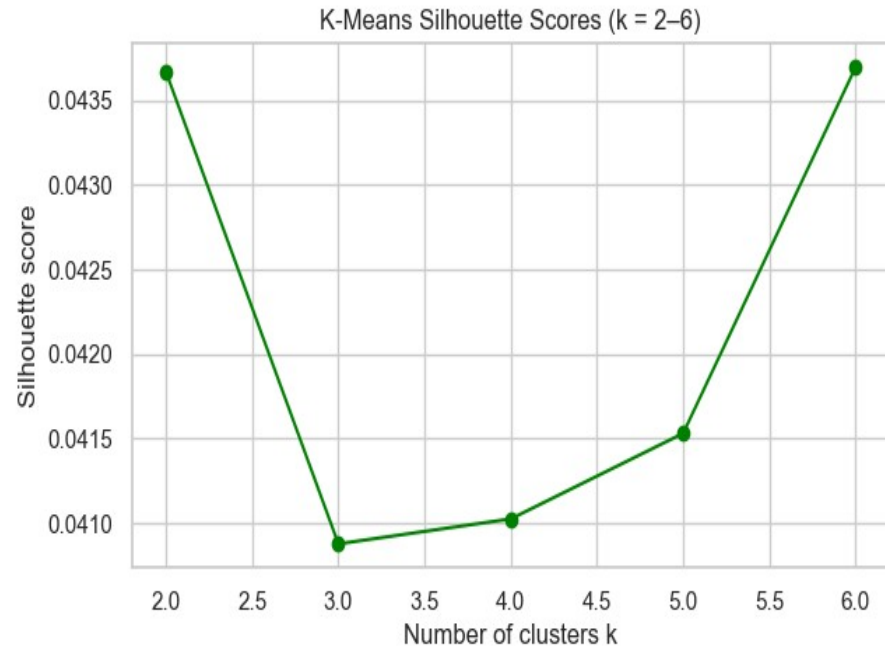
Features Included

- heart_rate, respiratory_rate, oxygen_saturation
- body_temperature, systolic_bp, diastolic_bp
- blood_glucose, muscle_oxygen, lactate_threshold
- workload, fatigue_index, recovery_time
- endurance_score, agility_score, reaction_time
- strength_score, hydration_level

K-Means: Elbow Method & Silhouette Analysis



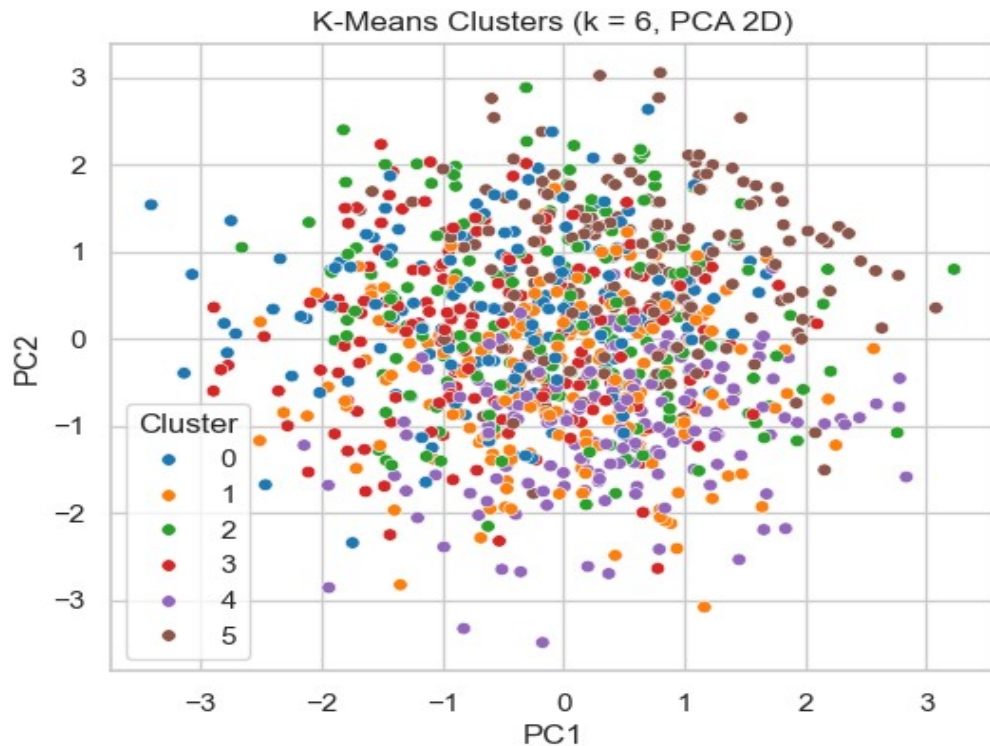
Elbow Curve (Inertia vs k)



Silhouette Score vs k

Selection: k = 6 chosen — silhouette peaks at k=2 and k=6; k=6 provides most interpretable performance profiles. *No sharp elbow visible — data forms a continuous spectrum of states rather than distinct clusters.*

K-Means Final Model (k = 6)



K-Means Clusters (k=6, PCA 2D)

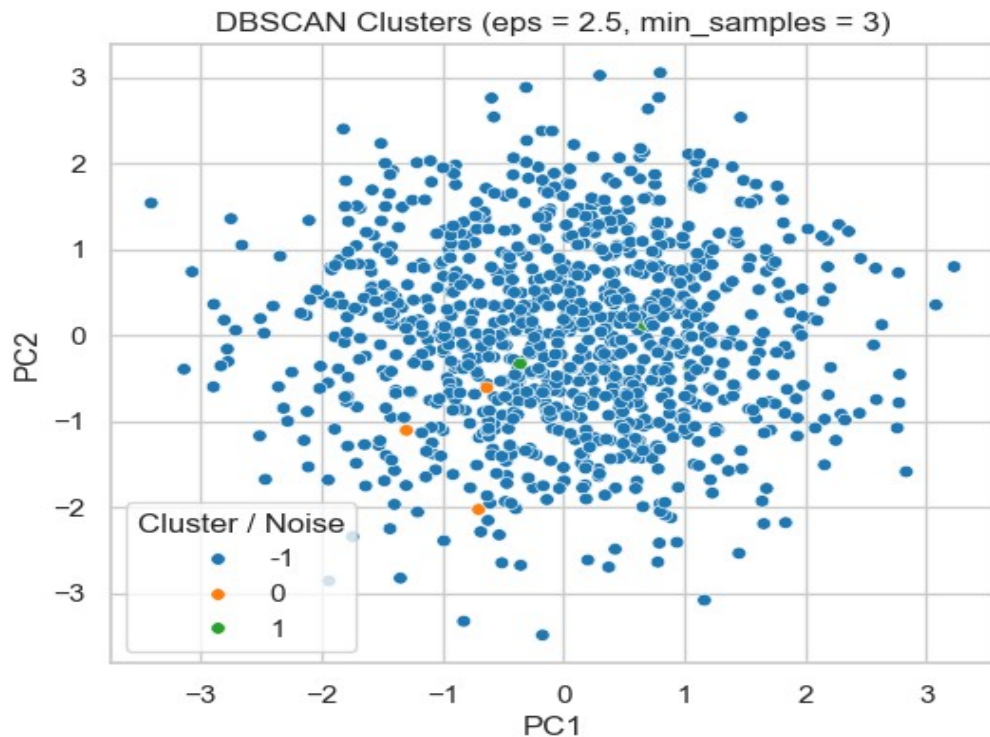
Results

Algorithm	K-Means
Clusters	6
Silhouette Score	Low (~0.04)
Inertia trend	Decreasing, no elbow

Key Observations

- Clusters heavily overlap in PCA 2D space
- Data forms a continuous spectrum, not distinct groups
- Clusters interpreted as soft performance profiles
- Tried k=2-7 and feature subsets — all gave low scores

DBSCAN: Parameter Search & Clustering



DBSCAN Clusters (eps=2.5, min_samples=3)

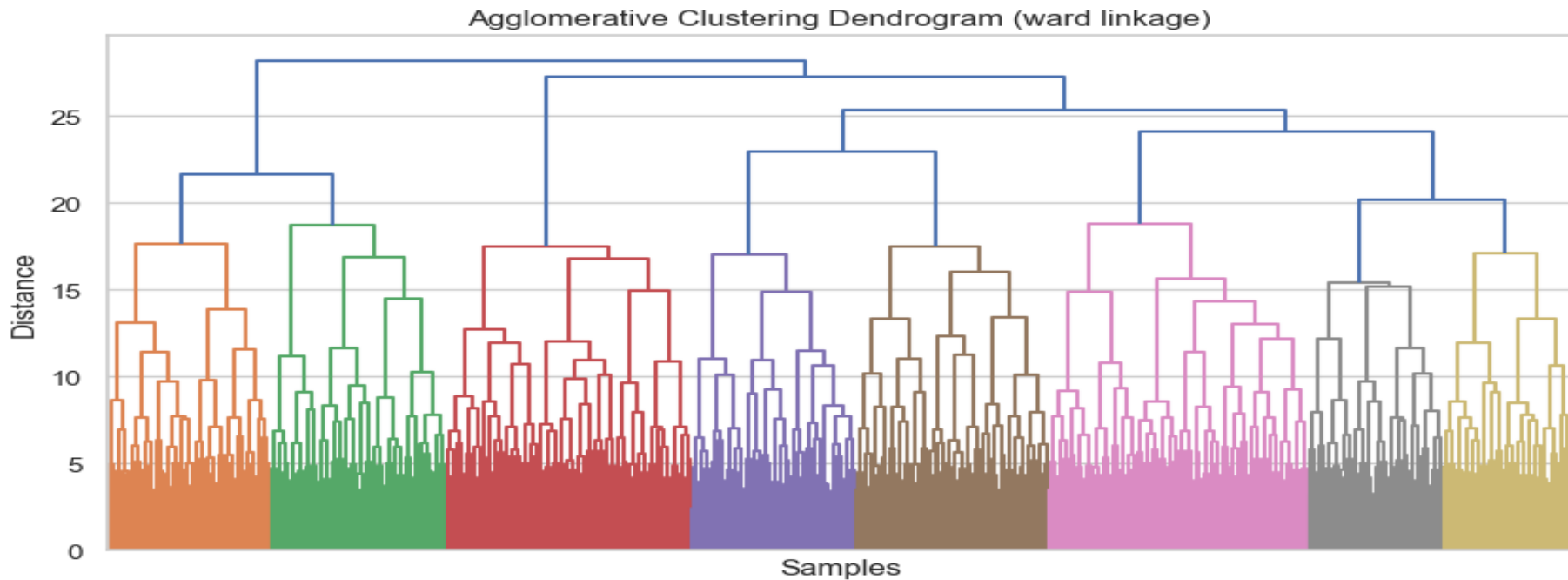
Best Parameters & Results

eps	2.5
min_samples	3
Clusters found	2 (very small)
Noise points	~995 (most points)
Silhouette Score	Negative

Interpretation

DBSCAN cannot identify meaningful clusters in this dataset. The data has no dense, well-separated regions — nearly all 1,000 points are classified as noise regardless of parameter settings.

Agglomerative Clustering: Dendrogram & Linkage



Agglomerative Dendrogram (ward linkage)

Linkage Comparison:

Dendrogram grows smoothly with no clear cut point — sessions form a continuous range of states.

Ward

0.013 ★ Best

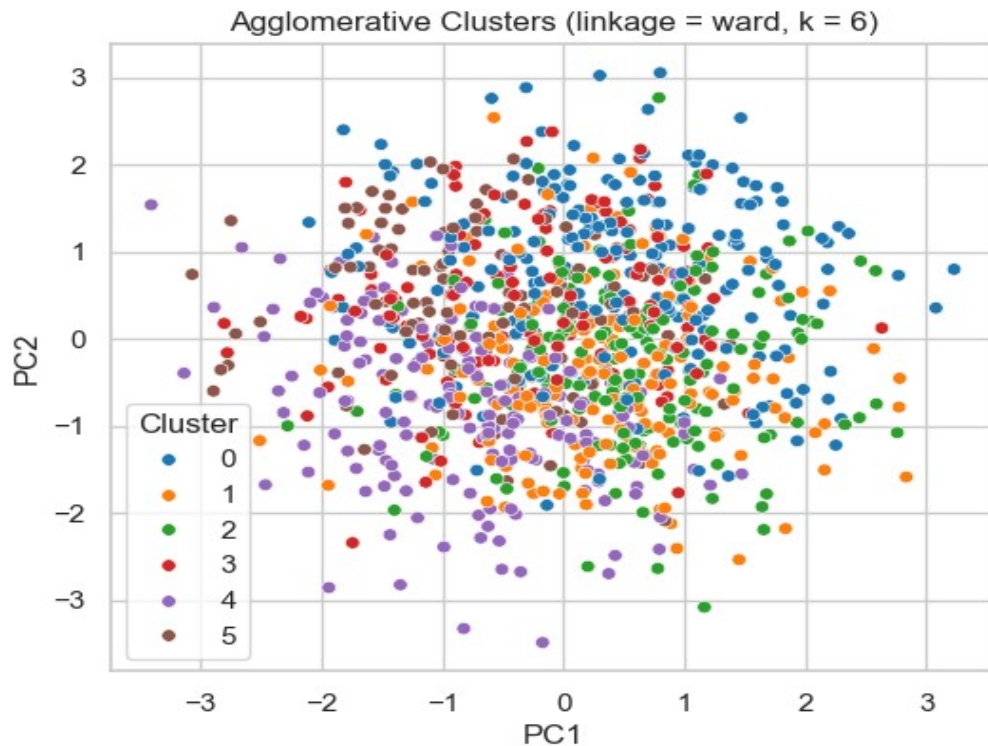
Complete

Lower

Average

Lower

Agglomerative Clustering — Final Result (Ward, k = 6)



Agglomerative Clusters (linkage=ward, k=6, PCA 2D)

Results

Algorithm	Agglomerative
Linkage	Ward (best)
Clusters	6
Silhouette Score	0.013

Key Observations

- Very similar structure to K-Means
- High overlap between clusters in PCA space
- Ward linkage outperforms complete & average
- No clear hierarchical splits in the dendrogram
- Silhouette very close to zero — weak clusters

Algorithm Comparison Table

Algorithm	Clusters	Silhouette	Key Observations
K-Means	6	~0.04 ★	Weak, overlapping clusters; soft performance profiles; most interpretable result
DBSCAN	2	Negative	~995 points classified as noise; no useful dense groups found
Agglomerative (Ward)	6	0.013	Similar overlap to K-Means; no clear hierarchical splits; smooth dendrogram

Recommendation

K-Means with k=6 is selected as the preferred method. While all algorithms produced low silhouette scores — confirming the data lacks strongly separated clusters — K-Means provides the most stable and interpretable groupings. The six clusters are best understood as soft 'performance profiles' representing a continuous spectrum of athlete physiological states.

Conclusion

3 Algorithms Applied

K-Means (k=6)
DBSCAN (eps=2.5, min=3)
Agglomerative (Ward, k=6)

Best Algorithm — K-Means

Most stable & interpretable result
6 soft performance profiles identified
Silhouette ~0.04 (best among three)

Key Challenge

Dataset forms a continuous spectrum
All algorithms gave low silhouette scores
No clear, well-separated clusters exist

Key Finding

Athlete physiological states are gradual
PCA reveals a single dense overlapping cloud
K-Means profiles useful for sports monitoring